

Sepehr Mahmoudian

SENIOR AI ENGINEER

Machine Learning • Knowledge Graphs • Multimodal & Embodied AI

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GITHUB github.com/sepahead

LINKEDIN linkedin.com/in/sepehr-mahmoudian

Berlin-Brandenburg • available from Nov 2026 • German citizen • EN fluent / DE good / Persian native



PROFILE

Builds custom agent harnesses for domain-specific applications, knowledge graphs and Graph RAG; evaluates LLMs, VLMs, VLAs and world models, pairing agentic orchestration with rigorous agentic engineering.

LLMs / VLMs • Knowledge Graphs • Agentic Vision • Multimodal AI • Scene Reconstruction
CONTEXT ENGINEERING • DATA PROVENANCE • MODEL EVALUATION • REINFORCEMENT LEARNING

AI ENGINEERING

10+ YEARS

RESEARCH TO PRODUCTION

PYTHON • RUST • AI / ML

PROFESSIONAL EXPERIENCE

Independent AI/ML Engineer & PhD Candidate

2020.04 - Present

Self-employed - PhD candidate at Technische Universität Darmstadt

Berlin & Darmstadt, Germany

- Building biophysical/functional neural models, PID (Partial Information Decomposition), Vision-Language-Action systems, VLA evaluation and neural control
- Building custom AI agent harnesses for domain-specific applications; multimodal 3D perception and auditable visualization

Senior AI Engineer

2026.01 - 2026.03

Neural Frames UG

Berlin & Hamburg, Germany

- Evaluated LLMs, VLMs and end-to-end pipelines, tracing runs and results with Langfuse
- Made limited contributions to Python pipelines for model evaluation and inference

AI Engineering Resident

2025.09 - 2026.01

Merantix AI Campus (Hacker Room)

Berlin, Germany

- Built generative-media systems with Gaussian splatting and diffusion models for spatiotemporal anchoring
- Implemented confidential-computing smart contracts in Rust
- Deployed product services and data infrastructure across Cloudflare Workers, D1, R2, Google Cloud Run and AWS EC2

Senior AI Engineer

2024.07 - 2025.07

Vision Labs UG

Berlin & Hamburg, Germany

- Built workflows for prompt optimization, LLM evaluation and observability with DSPy and Langfuse
- Designed data and MLOps pipelines on Azure and Google Cloud with staging, automated tests and CI/CD

Research Associate

2021.07 - 2024.07

Freie Universität Berlin

Berlin, Germany

- Scaled and optimized large neural-network simulations in C/C++ and Python
- Built retrieval-augmented generation (RAG) workflows for technical literature and evaluated AI-agent reliability
- Applied statistical and machine-learning analysis in Python; reviewed code and research manuscripts in an international team

Research Associate

2019.07 - 2020.06

Georg-August-Universität Göttingen • Göttingen, Germany

Neural networks, Partial Information Decomposition (PID), information theory and reproducible data workflows

Research Associate

2017.03 - 2019.07

Goethe-Universität Frankfurt • Frankfurt am Main, Germany • Simulation and statistical analysis of neural-network models

Visiting Scientist

2017.03 - 2019.03

Max Planck Institute for Dynamics and Self-Organization • Göttingen, Germany • Statistical methods for neural-network analysis

Research Associate

2015.02 - 2017.03

Forschungszentrum Jülich • Jülich, Germany • Neural-network simulations and reinforcement-learning experiments

Software Developer

2010.09 - 2012.03

Centre for Content Creation • Cyberjaya, Malaysia • Award-winning iPhone/iPad applications and digital products

EDUCATION

PhD Candidate (Dr. rer. nat.)

2020.04 - Present

Technische Universität Darmstadt • Darmstadt, Germany

• New mathematical PID analyses and information-theoretic representation analysis; Engram’s domain-specific AI agent produces provenance-tracked neural-model candidates for closed-loop neural-network simulations, with bounded compilation and simulation checks, in reproducible computational-neuroscience workflows

Master’s by Research in Biosciences: Neuroscience

2012 - 2014

University College London • London, United Kingdom

Computational Neuroscience • Graduated with Distinction (Top 10%, CGPA 4.0/4.0)

Bachelor of Science (Honours) in Computer Science

2009 - 2012

Anglia Ruskin University • Cambridge, United Kingdom

Graduated with First Class Honours (Top 1%, CGPA 4.0/4.0)

SELECTED AWARDS & FUNDING

EDTH / 2025

Special Mention at the European Defense Tech Hackathon Berlin 2025; co-organized with the Bundeswehr Cyber Innovation Hub

HUMAN BRAIN PROJECT

Research funding

NEAR HORIZON

Confidential-computing grant

UCL & SHEFFIELD

Competitive academic scholarships

IOS APPLICATIONS

15 international product awards

SIGNATURE

Sepehr Mahmoudian

PLACE Berlin, Germany

DATE 26 August 2026

PUBLIC ENGINEERING PORTFOLIO

GITHUB Open-source projects

github.com/sepahead

HUGGING FACE Models & datasets

huggingface.co/torusprime



SELECTED PROJECTS



engram

★ 15 RESEARCH PROTOTYPE

PROVENANCE-CONSTRAINED NEURAL R&D

Domain-specific AI agent turns qualified evidence into neural-model candidates, with bounded compilation and simulation checks.

Neural Networks Knowledge Graphs Closed-loop Sim



pid-rs

★ 14 V0.9 SOURCE PRERELEASE

VERIFIABLE INFORMATION DECOMPOSITION

Measures how much information several signals share about an outcome, with built-in checks and provenance that keep results reproducible.

Rust / Python Information Theory Statistical ML



prisoma

★ 2 V0.9 PRERELEASE / ACTIVE R&D

AUDITABLE CLOSED-LOOP EXPERIMENTS

Simulation environment for closed-loop VLA and world-model experiments, with capture, replay and provenance.

Rust / Python VLA + World Models Embodied Sim



NCP

★ 16 V0.8 RELEASE / 1.0 RC BLOCKED

NEURAL CONTROL PROTOCOL

Links neural simulators, AI controllers and robots through four typed planes with local fail-safes and provenance.

Rust Typed Control Planes Zenoh



manwe

★ 3 ALPHA

AIRSPACE PERCEPTION

Research workbench for vision, acoustics, multi-camera geometry and tracking; model exports must pass artifact-identity and numerical gates.

Python PyTorch Multimodal Perception



crebain

★ 21 RESEARCH PROTOTYPE

SPATIAL SITUATIONAL AWARENESS

Unifies Gaussian-splat scenes, native object detection and multimodal 3D tracking in one tactical view for simulated drone operations.

Rust Sensor Fusion Native Inference



melkor

★ 13 HARDENING

GAUSSIAN-SPLAT INTEGRITY

Inspects and deterministically converts 3D Gaussian splats, surfacing numeric hazards and field provenance before pipeline use.

C++ Scene Reconstruction Asset Validation



cobot-atlas relief-atlas

★ 7 +1,850 PUBLISHED + DOI

AI-GENERATED SIMULATION DATASETS

2,023 robot-simulation and 10,079 disaster-relief GLB meshes with generation pipelines.

Python Hugging Face 3D / VLA



galadriel

★ 7 RESEARCH PROTOTYPE

CROSS-SENSOR CONSISTENCY

Fail-closed monitor for sensor disagreement on one track, using per-channel innovation tests and signed correlation over producer-declared projections.

Rust NIS / CUSUM Signed Correlation



cortexel

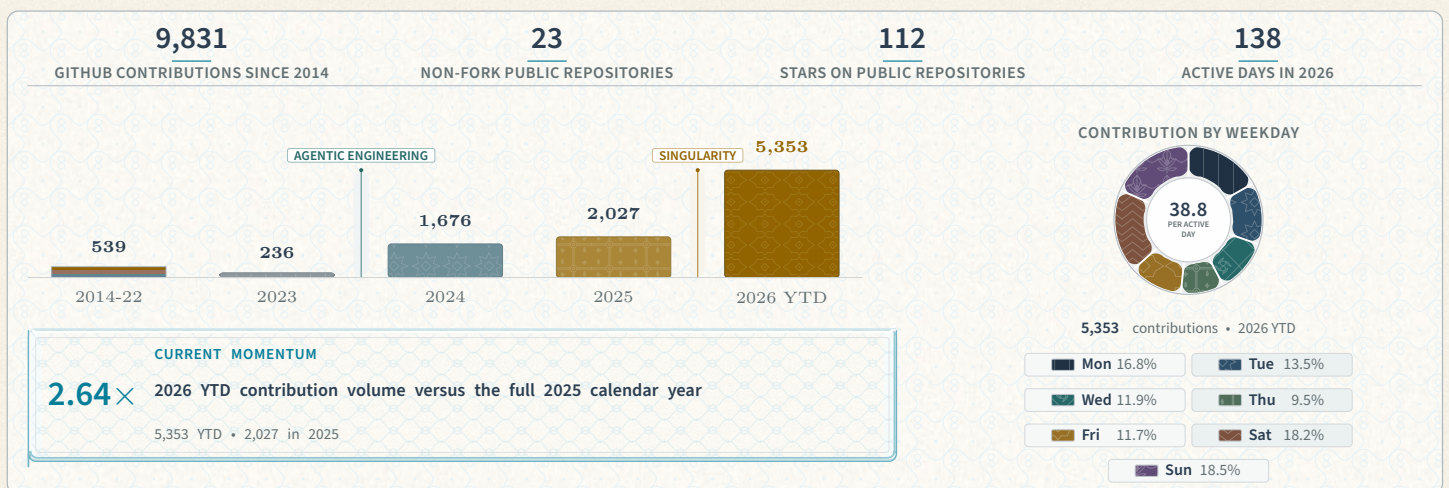
★ 4 0.10.0-DEV.0

AUDITABLE NEURAL FIGURES

Checks scientific figures against declared contracts before rendering, so charts provably match their data and stay reproducible.

Figure Contracts Deterministic SVG Provenance

PORTFOLIO & ENGINEERING ACTIVITY



TECHNICAL SKILLS

AI / ML ENGINEERING

• AGENTIC AI	LLMs / VLMs Agent Harnesses AI Agents Tool Use Model Evaluation Inference Pipelines
• MACHINE LEARNING	Python PyTorch Agentic Vision Multimodal AI Vision-Language-Action / VLA Evaluation
• KNOWLEDGE & RETRIEVAL	Knowledge Graphs Graph RAG RAG Embeddings Vector Search Reranking Data Provenance
• SYSTEMS & MLOPS	Rust C++ FastAPI ONNX TensorRT CUDA Metal / Core ML Docker
• CLOUD & INFRASTRUCTURE	GCP AWS Azure Kubernetes Terraform GitHub Actions Linux Zenoh

github.com/sepahead

huggingface.co/torusprime • models & datasets

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